THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** autotrophs	Auto means self.
2	**(a)** straight lines	Rectilinear propagation.
3	**(b)** west to east	Counter-clockwise from above the North Pole.
4	**(d)** blue-black	Starch test.
5	**(a)** a source of light, an opaque object and a screen	All three.
6	**(d)** 21 June	Summer solstice.
7	**(b)** leguminous plants	They fix nitrogen.
8	**(b)** reflection	Reflection.
9	**(b)** 27 days	Orbital period.
10	**(a)** make food for all	Base of the food chain.
11	**(b)** virtual and erect	Cannot be on a screen.
12	**(a)** Moon	Moon blocks Sun.
13	**(d)** photosynthesis	Photo means light.
14	**(a)** luminous objects	E.g. the Sun.
15	**(b)** 24 hours	It causes day and night.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Making of food by plants using sunlight (1); Carbon dioxide + water gives glucose + oxygen (1).	1+1
17	Luminous objects emit light (1); Non-luminous objects only reflect it (1).	1+1
18	Rotation: spin on its axis (1); Revolution: motion around the Sun (1).	1+1
19	Xylem carries water and minerals (1); Phloem carries food (1).	1+1
20	A box with a tiny hole and a screen (1); Forms an inverted image (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Carbon dioxide + water (1); In sunlight with chlorophyll (1); gives glucose + oxygen (1).	1+1+1
22	Look through three aligned cards with holes (1); Light seen when holes are aligned (1); Not when one is shifted (1).	1+1+1
23	Earth rotates (1); Side facing Sun has day (1); Other side has night (1).	1+1+1
24	Leaves form a pitcher (1); Insects fall in (1); Enzymes digest them (1).	1+1+1
	OR: Image 2 m behind (1); Total 4 m (1); Equal distances (1).	1+1+1
25	Earth between Sun and Moon (1); Earth's shadow on the Moon (1); At full moon (1).	1+1+1
	OR: Fix nitrogen (1); Enrich soil (1); Reduce fertiliser need (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Carbon dioxide and water (1); Sunlight (1); Chlorophyll in chloroplasts (1); Glucose and oxygen (1); In leaves (1).	1+1+1+1+1
	OR: Source, object, screen (1); Shadow forms behind (1); Moving the object closer to source enlarges (1); Moving closer to screen shrinks (1); Light travels straight (1).	1+1+1+1+1
27	Rotation: day and night (1); 24 hours (1); Revolution: year (1); 365.25 days (1); Seasons with tilt (1).	1+1+1+1+1
	OR: Take in oxygen (1); Give out carbon dioxide (1); Roots respire in soil air (1); Shed leaves (1); Store gums, resins (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** xylem	Xylem.
2	**(d)** closer to the light source	It blocks more light.
3	**(d)** 22 December	Winter solstice.
4	**(c)** fixing atmospheric nitrogen	Provide nitrates.
5	**(b)** a mirror	Polished surface.
6	**(d)** 29.5 days	Lunar month.
7	**(c)** plants	Plants make food.
8	**(a)** same as the object	Equal.
9	**(d)** new moon day	Aligned.
10	**(d)** chlorophyll	It traps sunlight.
11	**(a)** the Sun	It produces light.
12	**(b)** 365 and a quarter days	One year.
13	**(a)** phloem	Phloem.
14	**(d)** umbra	Complete shadow.
15	**(d)** 21 March and 23 September	Equinoxes.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	21 June (1); 22 December (1).	1+1
17	Two organisms live together to benefit each other (1); Lichen or Rhizobium with legumes (1).	1+1
18	To see over obstacles (1); Used in submarines (1).	1+1
19	At new moon (1); Moon between Sun and Earth (1).	1+1
20	Carbon dioxide (1); Water and sunlight (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Source is large (1); Complete shadow: umbra (1); Partial shadow: penumbra (1).	1+1+1
22	Orbit of Moon is tilted (1); Alignment is needed (1); Occurs rarely (1).	1+1+1
23	Cover part of a leaf (1); Test for starch after some hours (1); Only exposed part turns blue-black (1).	1+1+1
24	Shadow gets smaller (1); It becomes sharper (1); Source and screen fixed (1).	1+1+1
	OR: Axis tilted (1); Earth revolves around Sun (1); Different amounts of sunlight (1).	1+1+1
25	Alga makes food (1); Fungus gives shelter and water (1); Both benefit (1).	1+1+1
	OR: Two mirrors at 45° (1); Light reflected twice (1); Observer sees above obstacle (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Three mirror strips in a triangle (1); Coloured pieces at one end (1); Multiple reflections (1); Symmetric patterns (1); Rotating changes patterns (1).	1+1+1+1+1
	OR: No atmosphere (1); Craters (1); Weaker gravity (1); Extreme temperatures (1); No water seen on surface (1).	1+1+1+1+1
27	Variegated leaf test (1); Covered leaf test (1); Leaf in alkali (CO_2 absorbed) (1); Iodine test (1); Only suitable leaf parts give starch (1).	1+1+1+1+1
	OR: Light from top goes down (1); Light from bottom goes up (1); Rays cross at the hole (1); Image inverted (1); Small hole gives sharp image (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** symbiosis	Both benefit.
2	**(a)** incident ray	Incident.
3	**(d)** new moon	Amavasya.
4	**(c)** photosynthesis	Plants release.
5	**(d)** equal to	Equal behind the mirror.
6	**(a)** Earth	Earth's shadow falls on the Moon.
7	**(b)** carbon dioxide and water	Plus sunlight.
8	**(d)** the Moon	It reflects sunlight.
9	**(b)** rotation of the Earth	Half of the Earth faces the Sun.
10	**(d)** transpiration	Through stomata.
11	**(b)** penumbra	Partial shadow.
12	**(b)** 23.5° N	Northern boundary of the tropics.
13	**(b)** stomata and roots	Gas exchange.
14	**(b)** reflected ray	Reflected.
15	**(a)** full moon	Purnima.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Transparent: light passes fully (1); Translucent partly, opaque not at all (1).	1+1
17	Earth rotates (1); Half faces the Sun (1).	1+1
18	Parasitic (1); Saprotrophic (1).	1+1
19	Left and right are interchanged (1); In a plane mirror (1).	1+1
20	Day and night equal (1); 21 March and 23 September (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Year is 365.25 days (1); Quarter days add up (1); Extra day every 4 years (1).	1+1+1
22	Photosynthesis makes food (1); Respiration releases energy (1); One in light, other always (1).	1+1+1
23	It is non-luminous (1); Reflects sunlight (1); Seen by reflection (1).	1+1+1
24	Solar: Moon blocks Sun (1); Lunar: Earth's shadow on Moon (1); Solar at new moon, lunar at full moon (1).	1+1+1
	OR: Root hairs absorb (1); Xylem carries upward (1); Transpiration pulls water (1).	1+1+1
25	Box with a small hole on one side (1); Translucent screen on the opposite side (1); Inverted image forms (1).	1+1+1
	OR: Hemisphere tilted towards the Sun (1); Sun above horizon for longer (1); Opposite in winter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Axis tilted 23.5° (1); June: northern hemisphere tilted to Sun (1); Longer days, direct rays (1); December: southern tilted (1); Reverse seasons (1).	1+1+1+1+1
	OR: Make food (1); Release oxygen (1); Absorb CO2 (1); Give habitat (1); Prevent soil erosion (1).	1+1+1+1+1
27	Sun, candle, bulb, stars (1); Moon, book, table (1); We see by reflected light (1); Light enters the eye (1); Need light source (1).	1+1+1+1+1
	OR: Sun rises in east (1); West to east rotation (1); 24 hours for 360° (1); Different times in places (1); Time zones (1).	1+1+1+1+1

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